R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What are the types of security attacks?	[2]
b)	What are the two basic functions used in encryption algorithms?	[3]
c)	What are the advantages of public key cryptography algorithm?	[2]
d)	Define Stream ciphers.	[3]
e)	What is Hash function?	[2]
f)	What is meant by Kerberos?	[3]
g)	What are S/MIME message?	[2]
h)	Give the features of Authentication Header.	[3]
i)	What is secured electronic transaction?	[2]
j)	List the types of firewall.	[3]

PART - B

(50 Marks)

2.a) Explain the types of attacks in detail.
b) Describe in detail about Conventional Encryption Model. [5+5]

OR

3.a) Discuss about steganography.
b) Explain the transposition technique. [5+5]

4.a) Explain RSA algorithm.
b) In a RSA system, it is given that p = 11, q = 13, e = 7 and M = 5, Find ciphertext C and also M from decryption. [5+5]

OR

5. Explain DES algorithm with suitable examples. Discuss its advantages and limitations. [10]

6.a) Explain whirlpool algorithm.
b) Give various Hash Functions. Discuss secure hash algorithm with suitable examples. [5+5]

OR

7. Explain X.509 authentication service. [10]

8.a) Write a short note on Pretty Good Privacy.
b) Explain the IP Security architecture. [5+5]

OR

9.a) What do you mean by Security Association? Specify the parameters that identify the Security Association.
b) What are the cryptographic algorithms used in S/MIME? [5+5]

10. Explain Secure Socket Layer and Transport Layer Security. [10]

OR

11.a) Explain the design principles of firewalls.
b) Give an overview of virtual elections. [5+5]

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